

ACCV 2026 Reviews: Rebuttal Answers and Lessons for WACV 2027

29 August 2026

1 Purpose and evidence policy

This is a working research document, not yet a length-limited rebuttal. It records every reviewer question, the answer currently supported by the submitted results, and the work needed before reusing a claim in a WACV 2027 manuscript. Numerical claims below come from the ACCV branch tables. Items marked **Missing experiment/evidence** must not be answered with speculation or presented as established conclusions.

The central lesson is that the ACCV submission over-referenced appendix material that was not accessible with the main paper. Consequently, important evidence appeared to be missing even when files existed in the repository. Every future claim must be supported by a visible main-paper table/figure or by a correctly attached and resolvable appendix item.

2 Short factual summary

On the standard OccuFly test split, V-JEPA 2.1 FiLM-DPT-RGB obtains RMSE 4.621 m, AbsRel 0.129, δ_1 0.833, and SILog 0.164. DepthAnything2-OccuFly obtains RMSE 4.844 m, AbsRel 0.134, δ_1 0.834, and SILog 0.112. Thus the submitted model is better on RMSE, MAE, and AbsRel, essentially tied but slightly worse on the delta metrics, and substantially worse on SILog. The defensible conclusion is metric-error competitiveness, not overall superiority.

For SSC, the best submitted result is 37.68 SC IoU and 3.46 SSC mIoU. The best Pose-MVS result is 32.55 SC IoU and 2.36 SSC mIoU, below the simpler RGB-context result. These are negative but useful findings; they do not establish a new state of the art.

3 Reviewer-by-reviewer answers

3.1 Reviewer LD5V

3.1.1 LD5V-1: The paper lacks a clear end-to-end architecture/data-flow figure

Status: **Partial—claim/correction ready**

Answer. We agree. Figure 1 functions as an experiment map rather than an end-to-end architecture diagram, and the remaining figures do not make the connection among DTA, FiLM-DPT, Pose-MVS, depth-probability lifting, and SSC fusion sufficiently concrete. The intended flow is:

RGB/video + calibration + altitude $\rightarrow$ frozen V-JEPA 2.1
 $\rightarrow$ DPT/FiLM-DPT, DTA, or posed MVS branch
 $\rightarrow$ depth probabilities $\rightarrow$ 2D–3D lifting
 $\rightarrow$ SSC fusion and prediction head.

The revision should replace the screenshot-like overview with one legible diagram that shows tensor/data flow, frozen versus trainable modules, and the mutually exclusive depth branches. Each figure must be cited in the main text and checked at final PDF scale.

3.1.2 LD5V-2: Table 3 caption/format is inconsistent and claimed baselines are absent

Status: **Answered from existing evidence**

Answer. This is an authoring error. The caption claims MapAnythingV1.1, Metric3Dv2, Depth Anything 2, and Depth Anything 3, but the body contains only the DepthAnything2-OccuFly row. We will either restore all cited baseline rows using the same OccuFly protocol or revise the caption to name only the included baseline. Best values must be indicated per metric, not only for RMSE, and the discussion must cover all metrics.

3.1.3 LD5V-3: Table 4 has one method and cannot establish interpolation

Status: **Partial—claim/correction ready**

Answer. We agree that a single row is not a comparison. The existing number only shows that FiLM-DPT-RGB can be evaluated at held-out 40 m after training at 30/50 m; it does not isolate FiLM, nor does it establish scene generalization. The submitted prose must be weakened from “supports physically meaningful conditioning” to “a preliminary altitude-interpolation stress test.” A valid replacement table requires an otherwise identical unconditioned DPT control and preferably metadata ablations (altitude only, intrinsics only, altitude+intrinsics).

3.1.4 LD5V-4: Broken appendix references make evidence inaccessible

Status: **Answered from existing evidence**

Answer. We agree and apologize. The repository contains altitude/scene MDE tables and detailed SSC/per-class material, but the submitted package did not expose them through valid cross-references. This is not merely cosmetic because the missing items were the evidence for several claims. The revision must compile the main paper and appendix in one deliverable (or attach the supplement correctly), resolve every reference, and reject the build if “??” remains in the PDF or log.

3.1.5 LD5V-5: FiLM is not isolated because plain DPT is unexpectedly weak

Status: **Missing experiment/evidence**

Answer. The current results do not isolate the FiLM contribution. A DPT result that is worse than the linear probe is a warning of an optimization/configuration failure, not a sound control from which to measure a large FiLM gain. Before making a causal FiLM claim, we must audit preprocessing, frozen-layer selection, decoder initialization, depth loss, learning-rate/weight-decay, output scaling/bins, and checkpoint selection; rerun DPT and FiLM-DPT with identical settings and seeds; and add altitude-only and intrinsics-only ablations. Until then the defensible statement is that the *combined configured FiLM-DPT system* performs best among the tested V-JEPA heads.

3.1.6 LD5V-6: The altitude split is inconsistent and confounded by scene overlap

Status: **Answered from existing evidence**

Answer. The experiment report identifies train at 30/50 m, validation at 40 m scenes 01–04, and test at 40 m scenes 05–08. Therefore “scenes 1–9” is incorrect and must be corrected to scenes 01–08 with the validation/test partition stated explicitly. Because the same scene identities occur at other altitudes in training, the protocol holds out altitude but not scene identity. We will not call it scene-generalization evidence. A scene-disjoint altitude test is required for that claim.

3.1.7 LD5V-7: Video does not account for camera-motion misalignment

Status: **Missing experiment/evidence**

Answer. Correct. DTA aggregates tubelet tokens without explicitly warping features into the target camera frame. UAV ego-motion can therefore turn temporal aggregation into spatial mixing. The negative/altitude-dependent video result must be presented with this limitation. A controlled follow-up

should compare no alignment, pose-based feature warping, and (if available) optical-flow alignment using the same video decoder and training protocol.

3.1.8 LD5V-8: Pose-MVS is evaluated only through SSC, without direct depth results

Status: **Missing experiment/evidence**

Answer. We agree. Downstream SSC cannot diagnose whether Pose-MVS fails at correspondence/depth estimation or during lifting/fusion. We must report direct Pose-MVS depth metrics and qualitative depth/error maps on the same split as monocular depth, including an oracle-depth lifting upper bound. Until then, Pose-MVS is an exploratory negative SSC result, not evidence about MVS depth quality.

3.1.9 LD5V-9: SSC mIoU is low and lacks per-class, oracle, and statistical analysis

Status: **Partial—claim/correction ready**

Answer. Per-class and altitude-wise outputs exist in the repository but were made inaccessible by the appendix packaging failure. They must be restored and discussed, especially class imbalance and the gap between SC IoU and SSC mIoU. Oracle-depth lifting and multi-seed uncertainty are not present and must be added. The revised claim should be diagnostic: better monocular depth does not automatically translate into stronger semantic completion, and alignment between geometry, lifting, and context is a bottleneck.

3.2 Reviewer CWVJ

3.2.1 CWVJ-1: There is no second frozen backbone

Status: **Missing experiment/evidence**

Answer. We agree that a single-backbone study cannot attribute observed behavior specifically to V-JEPA 2.1. The current conclusion must be scoped to “V-JEPA 2.1 under this OccuFly protocol.” For WACV, at least one alternative frozen encoder—DINOv3 is the most directly motivated control raised by the reviewer—must use the same input resolution, selected feature levels, decoder capacity, training budget, splits, and seeds. If exact parity is impossible, parameter counts and deviations must be reported.

3.2.2 CWVJ-2: Swapping V-JEPA into SSC barely changes performance

Status: **Answered from existing evidence**

Answer. We accept this as a result rather than claiming a clear SSC gain. In the submitted configurations, V-JEPA-derived priors move SSC modestly and inconsistently; the architecture/lifting choice dominates. The contribution should therefore be framed as an empirical boundary: strong frozen-feature MDE transfer does not by itself guarantee SSC transfer.

3.2.3 CWVJ-3: Plain DPT appears broken

Status: **Missing experiment/evidence**

Answer. This is the same validity issue as LD5V-5. We cannot use the present DPT row as a clean causal baseline. The corrective experiment is a matched, multi-seed DPT versus FiLM-DPT comparison after a training/configuration audit, with learning curves and checkpoint-selection rules reported.

3.2.4 CWVJ-4: Broken appendix references and unfinished manuscript

Status: **Answered from existing evidence**

Answer. We agree. All evidence-bearing appendix items must ship with the paper, all labels must resolve, and tables/figures must remain readable in the final PDF. A clean compile check is part of the paper-release protocol below.

3.2.5 CWVJ-5: Mixed comparison to DepthAnything2 and confounded interpolation

Status: **Answered from existing evidence**

Answer. FiLM-DPT-RGB improves RMSE (4.621 vs. 4.844), MAE (3.523 vs. 4.193), and AbsRel (0.129 vs. 0.134), while it is marginally worse on δ_1 (0.833 vs. 0.834), δ_2 (0.971 vs. 0.976), and δ_3 (0.998 vs. 0.999), and markedly worse on SILog (0.164 vs. 0.112). This pattern suggests competitive absolute metric accuracy but less consistent scale-relative/log-space accuracy. It does not justify saying the method “beats DepthAnything2” without qualification. The interpolation limitation is addressed in LD5V-6.

3.2.6 CWVJ-6: Single runs and potentially seed-sized margins

Status: **Missing experiment/evidence**

Answer. We agree. Key comparisons require at least three controlled seeds, reported as mean $\pm$ standard deviation (and preferably paired per-scene differences or bootstrap confidence intervals). Claims based on margins comparable to uncertainty must be described as ties.

3.2.7 CWVJ-7: UAV efficiency motivation conflicts with an eight-H100 evaluation

Status: **Partial—claim/correction ready**

Answer. The paper reports the DGX hardware but no training/inference cost. Camera weight and power motivate visual sensing, but do not establish onboard deployability of a V-JEPA-Gigantic pipeline. We must remove or narrow any deployment implication and report trainable/total parameters, peak training memory, training GPU-hours, inference latency, throughput, and memory at the evaluated resolution. If the model is not tested onboard, state that explicitly.

3.2.8 CWVJ-8: Missing related work

Status: **Missing experiment/evidence**

Answer. The related-work section must be expanded and the proposed experiments positioned against: frozen multi-backbone 3D probing (Probe3D, Lexicon3D), calibrated or intrinsics-aware depth (CAM-ConvS and Metric3D’s canonical-camera formulation), aerial depth/geolocalization datasets (WildUAV, UseGeo, MidAir, AerialMetric), and real-flight video/multi-elevation benchmarks (RuralScapes, DronesCapes, MESSI). Each citation and dataset property must be verified from the primary paper/project before inclusion; no unverified bibliographic entry should be added from memory.

3.2.9 CWVJ-9: Contradictory SSC contribution wording

Status: **Answered from existing evidence**

Answer. The method text says the goal is not a completely new SSC architecture, while a contribution bullet can be read as claiming one. The consistent claim is that we evaluate V-JEPA-derived geometry/context variants within CGFormer/FoundationSSC-style pipelines, including an exploratory Pose-MVS branch. We will remove the new-architecture claim unless novelty is clearly defined and validated against matching controls.

3.3 Reviewer y7qY

3.3.1 y7qY-1: Explain why DepthAnything2 wins delta/SILog metrics

Status: **Answered from existing evidence**

Answer. The exact comparison is given in CWVJ-5. RMSE/MAE emphasize absolute metric errors (and RMSE strongly weights large errors), whereas delta accuracies measure the fraction within multiplicative thresholds and SILog measures log-space consistency. The observed trade-off is compatible with fewer/smaller large metric errors but a less uniform multiplicative error distribution. This is an interpretation, not a demonstrated mechanism; per-depth-range and error-distribution plots are needed to establish why.

3.3.2 y7qY-2: SSC is weak and video/MVS is a negative contribution

Status: **Answered from existing evidence**

Answer. We agree with the numerical characterization. The RGB-context configuration is strongest; video/MVS does not improve it. We will present this as a negative finding and investigate camera-motion alignment, direct MVS depth quality, and oracle-depth lifting before making a positive video-geometry claim.

3.3.3 y7qY-3: Broken references and figures not cited

Status: **Answered from existing evidence**

Answer. We will restore the missing appendix in the distributed package, repair all labels/references, cite every figure in the main text, and run the release checks listed below. Claims whose support does not fit in the main paper will explicitly point to a correctly attached supplementary item.

3.3.4 y7qY-4: Figure text may violate minimum font size

Status: **Partial—claim/correction ready**

Answer. All figures must be regenerated as vector graphics where possible and inspected at final single/double-column size. Labels will use at least the venue’s minimum font size. Screenshot-like tables should be replaced by native LaTeX tables or diagrams.

3.3.5 y7qY-5: Equation 19 says scenes 1–8 while Table 4 says 1–9

Status: **Answered from existing evidence**

Answer. Table 4 is erroneous. The recorded protocol uses 40 m scenes 01–04 for validation and 05–08 for testing, hence scenes 01–08, not 01–09. We will correct the table and state the overlap limitation.

4 Priority experiment plan before WACV 2027 claims

- P1. Repair the plain-DPT control and isolate conditioning:** matched DPT, altitude-only FiLM, intrinsics-only FiLM, and combined FiLM; same decoder, optimizer, budget, split, and at least three seeds.
- P2. Add a second frozen backbone:** DINOv3 (and optionally another strong encoder) under a capacity- and protocol-matched probe/decoder comparison.
- P3. Remove interpolation confounding:** create a scene-disjoint altitude protocol and compare FiLM with an unconditioned control.
- P4. Diagnose video geometry:** no alignment versus pose-warped/flow-aligned temporal features; report by altitude and motion magnitude.
- P5. Evaluate Pose-MVS directly:** standard depth metrics, qualitative errors, depth-bin calibration, and downstream SSC with predicted versus oracle depth.
- P6. Quantify uncertainty:** multi-seed mean/std and paired scene-level or bootstrap intervals for all headline comparisons.
- P7. Explain SSC failures:** accessible per-class and altitude tables, class frequencies, SC-versus-SSC analysis, and geometry/context/lifting controls.
- P8. Report computational cost:** parameters, GPU-hours, peak memory, latency, throughput, and a precise statement about whether onboard deployment was evaluated.

5 Paper-release checklist learned from ACCV

- Build the exact upload artifact from a clean checkout; do not assume local appendix files are included.
- Search the compiled source/log/PDF for unresolved references, citations, “??”, placeholder text, missing files, and overfull content.
- Cross-check every table caption against its actual rows, metrics, highlighting, split, units, and cited source.
- Ensure every figure/table is cited, legible at final size, and communicates a scientific point rather than serving as an experiment tracker.
- Keep a claim-to-evidence ledger: each abstract/contribution/conclusion claim maps to a main-paper or accessible supplementary table/figure.
- Distinguish observation, controlled ablation, and causal conclusion. Do not attribute an effect to FiLM, video, MVS, or V-JEPA without the corresponding control.
- Report negative results and uncertainty; do not promote a single-run numerical margin to a general conclusion.

6 Candidate WACV 2027 framing

The strongest defensible direction is not “a new SSC architecture.” It is a controlled, multi-backbone study of when frozen visual features transfer from aerial metric depth to semantic scene completion, and why that transfer can fail. The ACCV result supplies the initial observation: frozen V-JEPA 2.1 plus a conditioned dense decoder is competitive in absolute metric-depth errors, yet video/MVS and stronger depth do not automatically yield better SSC. The new WACV experiments should turn that observation into evidence by adding matched backbones, conditioning ablations, scene-disjoint generalization, motion-aware video controls, direct MVS evaluation, oracle lifting, and uncertainty.